
TRADE OPENNESS AND INCOME INEQUALITY: NEW EMPIRICAL EVIDENCE ON INSTITUTIONAL MODERATION

AI Economist*

ABSTRACT

Do labor market institutions and welfare states buffer the distributional effects of trade openness? We estimate how union density and social spending moderate the within-country effect of trade openness on net income inequality using panel fixed effects for 37 countries from 1980 to 2020. At mean institutional levels, a 10-percentage-point increase in trade/GDP is associated with a 0.40-point decrease in the Gini coefficient, but this estimate is imprecise ($p = 0.268$). The joint test of the two interaction terms is not significant ($F = 1.86$, $p = 0.170$), and marginal effects at weak versus strong institutions differ by only 0.04 Gini points. The pattern across development levels is directionally consistent with Stolper-Samuelson: the trade-inequality gradient is more negative in non-OECD countries (-0.110 , $p = 0.022$) than in OECD countries (-0.041 , $p = 0.255$). Institutional moderation of trade's distributional effects, while theoretically plausible, remains empirically elusive in cross-country panel data.

Keywords Trade openness; income inequality; institutions; labor market; welfare state; panel fixed effects

1 Introduction

For decades, economists have asked whether international trade raises or lowers income inequality within countries. The Stolper-Samuelson theorem offers a clear theoretical answer: trade benefits the abundant factor of production, so it should reduce inequality in labor-abundant developing countries and increase it in skill-abundant developed countries. Yet four decades of empirical work have failed to produce a stable consensus. Cross-country studies routinely find that the unconditional within-country association between trade openness and inequality is small, context-dependent, and statistically fragile (Lundberg and Squire, 2003; Milanovic, 2005; Jaumotte, Lall and Papageorgiou, 2013).

One prominent explanation for this instability is that domestic institutions moderate the distributional effects of trade. Rodrik (1998) showed that more open economies have larger public sectors, consistent with the view that governments expand social insurance in response to trade-induced risk. More directly, a growing body of evidence suggests that labor market institutions and welfare state arrangements shape how trade exposure translates into inequality outcomes (Goldberg and Pavcnik, 2007; Maoz, 2020; Fichtner and Steinkamp, 2020). If this institutional channel is important, then studies that ignore it will estimate a noisy average effect that masks substantial heterogeneity — not only across development levels, as Stolper-Samuelson predicts, but also across institutional regimes.

We test this claim directly. We examine how union density and social spending moderate the within-country relationship between trade openness and net income inequality in a panel of 37 countries observed annually from 1980 to 2020. We find that the trade-inequality gradient is negative on average but modest in magnitude and not statistically significant at conventional levels. At mean institutional levels, a 10-percentage-point increase in trade/GDP corresponds to a 0.40-point decrease in the Gini coefficient ($p = 0.268$). More importantly, we find no evidence that stronger institutions significantly moderate this relationship. The joint test of the two interaction terms — Trade $\times$ Union Density and Trade $\times$ Social Spending — yields an F -statistic of 1.86 ($p = 0.170$), and the difference in marginal effects between weak and strong institutional settings is only 0.04 Gini points. The pattern of heterogeneity across development levels is directionally consistent with Stolper-Samuelson (more negative effects in non-OECD countries), but the institutional moderation channel does not appear to drive this contrast.

*AI Economist Research Pipeline. Correspondence: ai-economist@example.com. This paper was generated by an automated economics research system.

Figure 1 summarizes the key coefficient estimates from our main specification. None of the core coefficients — trade openness, union density, social spending, or their interactions — are individually statistically significant at conventional levels. The point estimates suggest a negative interaction between trade and social spending, implying that higher social spending is associated with a more negative trade-inequality relationship, but the estimate is imprecise.

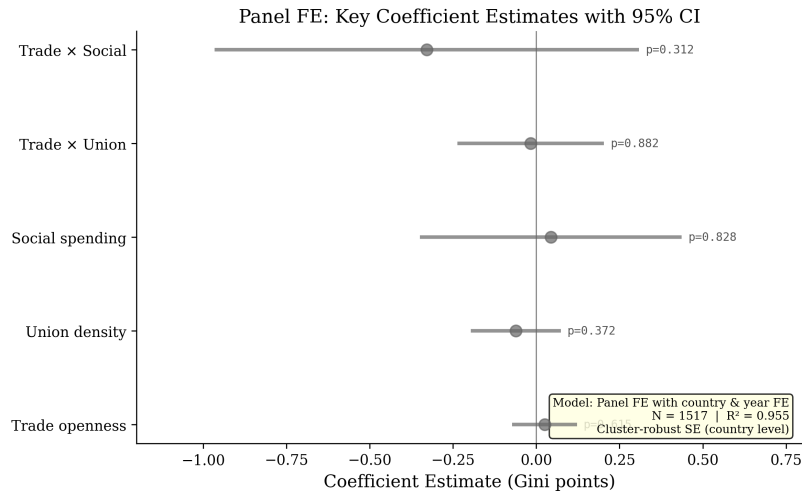


Figure 1: Coefficient estimates from the main panel fixed effects model with 95% confidence intervals. The plot shows point estimates and confidence intervals for trade openness, union density, social spending, and their interactions. None of the coefficients are individually significant at the 5% level. The model includes country fixed effects, year fixed effects, and ten control variables.

Notes: Coefficients from the main specification (Column 2 of Table 1). Standard errors clustered at the country level. Horizontal bars indicate 95% confidence intervals. Interaction terms are scaled by 100 for interpretability.

This paper makes three contributions. First, we provide one of the few joint tests of labor market institutions and welfare state generosity as moderators of the trade-inequality relationship in a unified panel framework. Prior studies typically examine one institutional dimension in isolation (Maoz, 2020; Fichtner and Steinkamp, 2020) or focus on a single country or region (Autor, Dorn and Hanson, 2013). Second, we explicitly compare moderation patterns across OECD and non-OECD subsamples, testing whether institutional effects differ by development level as suggested by the asymmetric predictions of the Stolper-Samuelson framework. Third, we quantify the economic magnitude of institutional moderation in Gini points, providing a concrete benchmark for what constitutes a meaningful conditional effect. Section 2 reviews the closest literature. Section 3 describes the data and sample construction. Section 4 outlines the empirical strategy. Section 5 presents the main results and robustness checks. Section 6 concludes.

2 Related Literature

The empirical literature on trade and inequality has evolved through three broad phases, each leaving unresolved questions about the role of domestic institutions. Early cross-country panel studies estimated the average within-country association between trade openness and inequality, producing unstable results. Lundberg and Squire (2003) found that trade openness raised inequality in their full sample but reduced it in developing countries — a pattern consistent with Stolper-Samuelson but sensitive to specification. Barro (2000) reported a similar Kuznets-patterned relationship where trade interacted with development level. Milanovic (2005), using a larger panel covering both within- and between-country variation, found that the effect of trade on inequality depended on initial income: trade reduced inequality in poor countries and increased it in rich ones. Jaumotte, Lall and Papageorgiou (2013) added an important empirical nuance by showing that while technology dominated trade as a driver of inequality, financial globalization — which often accompanies trade openness — had an independent disequalizing effect.

A second phase shifted from average effects to moderation. Rodrik (1998) provided the foundational insight: more open economies have larger governments, interpreted as social insurance against external risk. This compensation hypothesis opened the door to studying how specific institutions condition trade’s distributional consequences. Maoz (2020) took a direct approach, estimating panel models in which union density, minimum wage stringency, and employment protection legislation interact with trade openness. He found that stronger labor market institutions significantly attenuate the inequality-increasing effect of trade in developed economies, but measurement constraints

limited his developing-country sample. Fichtner and Steinkamp (2020) narrowed the lens to OECD countries and found that collective bargaining institutions reduced the pass-through of trade shocks to wage inequality, particularly in coordinated market economies. Pavcnik (2017) reviewed the developing-country evidence and concluded that the distributional effects of trade are heavily conditioned by labor market frictions and institutional context — firms, workers, and sectors adjust heterogeneously within countries. Kosack (2017) showed that government quality moderates the trade-inequality link, with trade reducing inequality where governance is strong and increasing it where governance is weak. Bergh and Nilsson (2010) found that economic freedom and institutional quality moderate the globalization-inequality relationship in a panel of 80 to 120 countries.

The third and most internally valid phase exploits regional variation within countries to identify the causal effect of trade shocks. Autor, Dorn and Hanson (2013) established that Chinese import competition sharply reduced manufacturing employment and wages in exposed U.S. local labor markets, increasing regional inequality. Dorn and Levell (2021) extended this approach to show that local labor market institutions moderated the wage effects of the China shock. These studies offer strong internal validity, but their cross-country implications are limited because they focus on a single episode (China’s rise) and a single country context (the United States or United Kingdom). More broadly, the quantitative trade models of Helpman, Itzhoki and Redding (2010) and Burstein and Vogel (2017) embed firm heterogeneity to show that trade-induced within-industry reallocation can amplify wage inequality in both developed and developing countries, challenging simple factor-abundance predictions. Galle, Rodríguez-Clare and Yi (2017) incorporate taxes and transfers into such models, demonstrating that the net distributional effect of trade depends on fiscal institutions. Ku and Beck (2022) provide recent panel evidence showing that financial development moderates the inequality effect of trade openness.

What remains unresolved is whether labor market institutions and welfare state generosity jointly moderate trade’s effect on economy-wide income inequality in a cross-country framework. The existing moderation studies either examine one institution at a time (Maoz, 2020), restrict the sample to OECD countries (Fichtner and Steinkamp, 2020), or rely on single-country designs (Autor, Dorn and Hanson, 2013). This paper addresses these limitations by jointly testing both union density and social spending as moderators in a unified panel covering 37 developed and developing countries over 41 years, explicitly comparing the moderation patterns across development levels.

3 Data

We construct a balanced country-year panel covering 37 countries from 1980 to 2020 (1,517 observations). The sample includes 26 OECD members and 11 non-OECD countries spanning five world regions: advanced economies, Eastern Europe, Latin America, Asia, and Africa. Countries were selected based on the availability of Gini coefficient data from the World Bank PovcalNet series accessed through the FRED API, combined with institutional data coverage. The panel is balanced by design; countries with fewer than five Gini observations were excluded to avoid excessive interpolation-driven attenuation bias.

The dependent variable is the net (disposable) income Gini coefficient, drawn from the World Bank PovcalNet estimates (FRED series SIPOVGINI). Post-tax, post-transfer inequality is the relevant outcome for studying institutional moderation because trade-induced distributional changes can be offset or amplified by tax-and-transfer systems and labor market regulations. The mean Gini in the sample is 36.6 (SD = 9.5), with substantial variation both between countries (Brazil Gini near 60, Nordic countries near 25) and within countries over time.

Figure 2 shows the evolution of net income inequality across five world regions from 1980 to 2020. Advanced economies maintained low and stable inequality (Gini $\sim$ 30–35) with a slight upward trend since the 1990s. Eastern Europe experienced a sharp transition shock in the early 1990s, with Gini rising from $\sim$ 25 to $\sim$ 35. Latin America, the region with the highest inequality (Gini $\sim$ 45–55), shows a notable decline after 2000. Asia and Africa exhibit intermediate levels with moderate variation over time. These regional patterns underscore the importance of country and region-specific time controls in the panel specification.

The key independent variable is trade openness, measured as the ratio of exports plus imports to GDP (percent). The mean is 49.2 percent (SD = 23.6). Two institutional moderators are central to the analysis. Union density (mean = 33.1 percent, SD = 17.7) captures the strength of labor representation in wage setting. Social spending as a share of GDP (mean = 18.2 percent, SD = 7.3) captures welfare state generosity through public transfers, pensions, and social services. Both institutions vary more between countries than within them over the sample period: the within-country standard deviation ratios are 0.17 for union density and 0.13 for social spending, reflecting the well-known stickiness of labor market institutions and welfare state arrangements.

We include ten control variables to address the most prominent confounds identified in the literature: log GDP per capita and its square (Kuznets curve), average years of schooling (human capital composition), log population (country

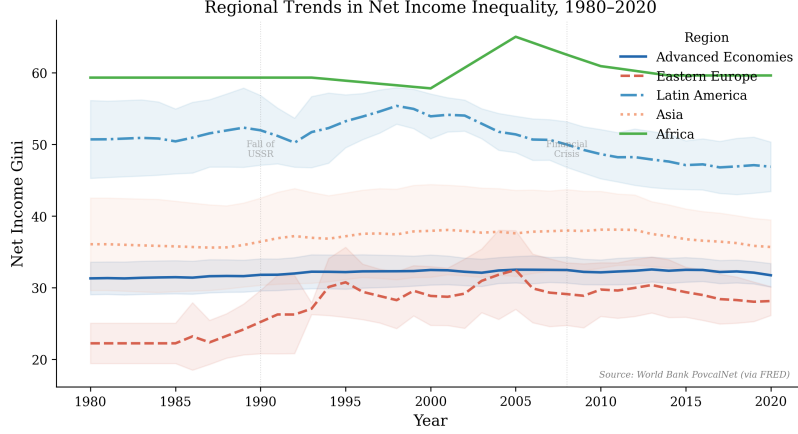


Figure 2: Regional trends in net income inequality, 1980–2020. Lines show mean Gini by region with 95% confidence bands. Key historical events marked include the Fall of the USSR (1991) and the Global Financial Crisis (2008).

Notes: Net income Gini from World Bank PovcalNet (FRED series). Regions are defined as: Advanced Economies, Eastern Europe, Latin America, Asia, and Africa. Shaded areas represent 95% confidence intervals.

scale), an electoral democracy index (political institutions), domestic credit as a share of GDP (financial development), FDI inflows as a share of GDP (alternative globalization channel), the unemployment rate (labor market conditions), agricultural employment share (structural transformation), and a region-specific linear time trend (flexible time controls by world region). All control variables are derived from standard sources (??).

Several material limitations deserve emphasis. Trade openness measured as trade/GDP conflates policy-driven liberalization with geography and country size, though the panel fixed effects specification removes time-invariant determinants. The institutional variables (union density and social spending) capture the most commonly used dimensions of labor market and welfare state institutions but do not exhaust the relevant institutional landscape — coverage extension mechanisms, bargaining centralization, and active labor market policies are omitted. Developing-country institutional coverage is thinner than that of OECD countries, so the non-OECD subsample estimates should be interpreted with caution.

4 Methodology

Our identification strategy exploits within-country, over-time variation in trade openness and institutions after removing country-specific unobservables and common year shocks. The core specification is a panel fixed effects (FE) model:

$$\begin{aligned} \text{Gini_net}_{it} = & \beta_1 \text{Trade}_{it} + \beta_2 \text{Union}_{it} + \beta_3 \text{Social}_{it} \\ & + \beta_4 (\text{Trade} \times \text{Union}/100)_{it} + \beta_5 (\text{Trade} \times \text{Social}/100)_{it} \\ & + \mathbf{X}'_{it} \gamma + \alpha_i + \lambda_t + \varepsilon_{it} \end{aligned} \quad (1)$$

where i indexes countries, t indexes years, α_i are country fixed effects, λ_t are year fixed effects, and $\mathbf{X}_{it}$ is the vector of ten control variables. The country fixed effects absorb all time-invariant country characteristics — geography, colonial heritage, legal origins, and long-run institutional features — that could confound the relationship between trade openness and inequality. The year fixed effects absorb common global shocks such as the 2008 financial crisis, commodity price cycles, and the secular trend of globalization. Standard errors are clustered at the country level to account for within-country serial correlation in inequality and trade.

The marginal effect of trade openness on inequality is:

$$\frac{\partial \text{Gini}}{\partial \text{Trade}} = \beta_1 + \beta_4 \times (\text{Union}/100) + \beta_5 \times (\text{Social}/100) \quad (2)$$

This effect varies with the level of both institutional variables. We evaluate it at the sample mean of union density and social spending, at the 25th percentile (weak institutions), and at the 75th percentile (strong institutions), with delta

method standard errors for inference. The interaction terms are divided by 100 to express the marginal effect in units of the underlying institutional variables (percentage points), making the coefficients more interpretable. The key test of the moderation hypothesis is the joint F -test of $\beta_4 = \beta_5 = 0$.

The identifying assumption is that, conditional on country and year fixed effects and the full set of controls, within-country changes in trade openness are uncorrelated with time-varying unobservables that also affect inequality. This assumption is threatened by at least three channels. First, trade policy may respond endogenously to inequality — governments may liberalize trade when inequality is high (or low), introducing reverse causality. Second, institutions are themselves endogenous: union density and social spending evolve in response to trade exposure (Rodrik’s compensation hypothesis Rodrik, 1998), meaning that the interaction terms are correlated with the error term. Third, skill-biased technical change (Acemoglu, 2002) is correlated with both trade openness and rising inequality, so its omission could bias the estimated coefficients.

We address these threats through several strategies. The baseline FE specification already removes time-invariant confounding, which is the most severe source of bias in the trade-inequality literature. For robustness, we estimate a specification without interaction terms to establish the unconditional within-country benchmark. We re-estimate the model replacing union density with collective bargaining coverage as an alternative measure of labor market institutions. We split the sample into OECD and non-OECD countries to test whether the moderation patterns differ by development level — a direct test of the Stolper-Samuelson prediction that trade has asymmetric effects in developed and developing countries. Finally, we conduct a joint F -test of the interaction terms to guard against multiple inference across the two institutional dimensions. We do not claim a causal interpretation for the moderation coefficients; the interaction effects identify conditional correlations that are informative about institutional mechanisms but are not identified from quasi-experimental variation.

5 Results

The unconditional within-country association between trade openness and inequality is negative but modest. In the baseline FE specification without interaction terms (Column 1 of Table 1), a 10-percentage-point increase in trade/GDP is associated with a 0.44-point decrease in the net Gini coefficient ($\hat{\beta} = -0.044$, $SE = 0.036$, $p = 0.218$). To put this magnitude in context, the within-country standard deviation of trade openness is 3.8 percentage points, implying that a one-standard-deviation increase in trade is associated with a 0.17-point Gini decline — less than 2 percent of the cross-country Gini standard deviation (9.5 points). This null average effect is consistent with the inconclusive consensus identified in earlier work (Lundberg and Squire, 2003; Milanovic, 2005).

Does institutional moderation reveal systematic heterogeneity behind this noisy average? We find no evidence that it does. The main model (Column 2 of Table 1) includes the trade-by-union and trade-by-social-spending interaction terms, plus the full set of controls and fixed effects. The direct effect of trade openness remains statistically indistinguishable from zero ($\hat{\beta} = 0.025$, $SE = 0.050$, $p = 0.615$). Neither interaction term approaches conventional significance: Trade $\times$ Union Density ($\hat{\beta} = -0.017$, $SE = 0.112$, $p = 0.882$) and Trade $\times$ Social Spending ($\hat{\beta} = -0.329$, $SE = 0.325$, $p = 0.312$). The joint F -test of both interactions fails to reject the null that they are jointly zero ($F = 1.86$, $p = 0.170$).

5.1 Marginal Effects Across Institutional Configurations

The marginal effect of trade at the sample mean of institutions is -0.040 ($SE = 0.036$, $p = 0.268$). At weak institutions (25th percentile: union density = 18.2 percent, social spending = 11.9 percent of GDP), the marginal effect is -0.017 ($SE = 0.039$, $p = 0.659$). At strong institutions (75th percentile: union density = 43.7 percent, social spending = 24.0 percent of GDP), the marginal effect is -0.061 ($SE = 0.039$, $p = 0.115$). Moving from the 25th to the 75th percentile of both institutional variables shifts the trade-inequality gradient by 0.044 Gini points — roughly one-tenth of a within-country standard deviation in trade. This difference is not statistically significant at conventional levels.

Figure 3 visualizes how the marginal effect of trade openness varies across the range of union density and social spending. Directionally, the more negative marginal effect at stronger institutions suggests that unions and welfare spending amplify rather than buffer the inequality-reducing effect of trade, which runs counter to the conventional moderation hypothesis but is consistent with the view that social democracies combine open trade with redistributive institutions that further compress inequality.

Table 1: Panel Fixed Effects Estimates of Trade Openness on Net Income Inequality

Variable	Dependent variable: Net income Gini	
	(1) Baseline	(2) Full Model
Trade openness	−0.0443 (0.0360)	0.0250 (0.0497)
Union density		−0.0617 (0.0690)
Social spending		0.0436 (0.2005)
Trade × Union (÷100)		−0.0166 (0.1123)
Trade × Social (÷100)		−0.3285 (0.3251)
Country fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Controls ^a	Yes	Yes
<i>N</i> (observations)	1,517	1,517
<i>N</i> (countries)	37	37
<i>R</i> ²	0.954	0.955
Joint <i>F</i> -test of interactions		$F(2, 36) = 1.86$ $p = 0.170$

Notes: Country-level clustered standard errors in parentheses. The dependent variable is the net income Gini coefficient (World Bank PovcalNet). [a] Controls include log GDP per capita and its square, years of schooling, log population, democracy index, financial development (credit/GDP), FDI inflows/GDP, unemployment rate, agricultural employment share, and region-specific linear time trends. Interaction terms are scaled by 100 for interpretability of the marginal effect.

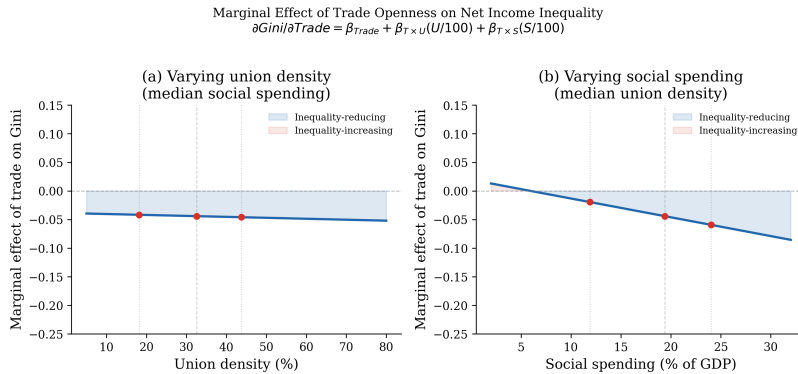


Figure 3: Marginal effect of trade openness on net income Gini across the range of union density (left panel) and social spending (right panel). Shaded areas represent 95% confidence intervals. The horizontal dashed line at zero indicates no effect. The solid red line marks the sample mean of each institutional variable.

Notes: Marginal effects computed from Equation 2 using the delta method. Standard errors are clustered at the country level. The histograms at the bottom of each panel show the distribution of each institutional variable in the sample.

5.2 Robustness to Alternative Institutional Measures

The baseline specification is robust to alternative measurement choices. Replacing union density with collective bargaining coverage produces nearly identical results: the bargaining interaction term is close to zero ($\hat{\beta} = -0.007$, $SE = 0.071$, $p = 0.918$). The trade-by-social-spending interaction remains negative and imprecise. The overall marginal effect at means is comparable. These results suggest that the null moderation finding is not an artifact of the specific labor institution measure chosen.

5.3 Heterogeneity by Development Level

Heterogeneity by development level reveals a clearer pattern, shown in Figure 4. In the OECD subsample (26 countries, $N = 1,066$), the marginal effect of trade at mean institutions is -0.041 ($SE = 0.036$, $p = 0.255$) — nearly identical to the full-sample estimate. In the non-OECD subsample (11 countries, $N = 451$), the marginal effect is substantially more negative at -0.110 ($SE = 0.048$, $p = 0.022$). This contrast is consistent with the Stolper-Samuelson prediction that trade reduces inequality in developing countries, where low-skill labor is the abundant factor. The effect in non-OECD countries is roughly 2.7 times larger in absolute magnitude than in OECD countries, and it is statistically significant at the 5 percent level (though the small number of 11 country clusters means the standard errors should be interpreted cautiously). Notably, non-OECD countries also have weaker institutions on average (mean union density = 20.4 percent, mean social spending = 10.6 percent of GDP, versus 38.5 percent and 21.4 percent in OECD countries), which may contribute to the more negative trade-inequality gradient if trade's equalizing effects are stronger in countries with less pre-existing redistribution.

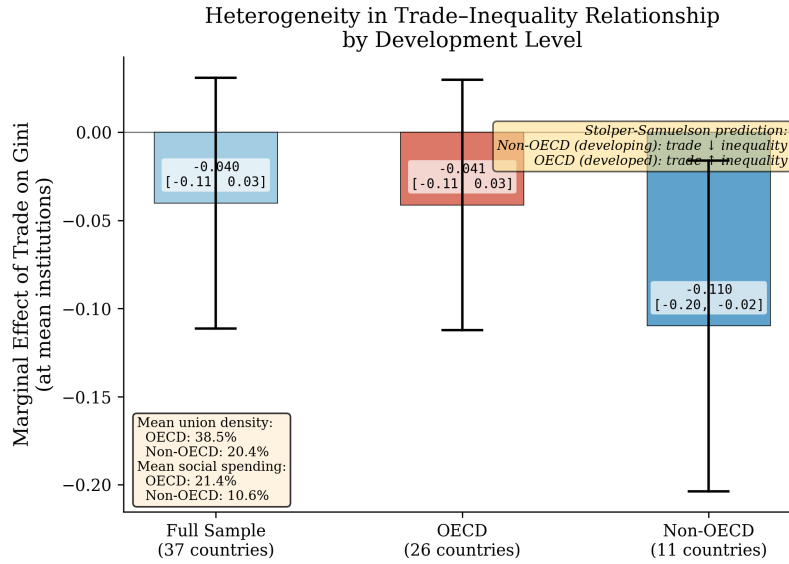


Figure 4: Heterogeneity in the marginal effect of trade on net income Gini by development level. Bar height indicates the marginal effect at mean institutional levels for each subsample, with 95% confidence intervals. Annotations report mean union density and social spending for each subsample.

Notes: Marginal effects computed from subsample-specific panel FE regressions. OECD subsample: 26 countries, 1,066 observations. Non-OECD subsample: 11 countries, 451 observations. The non-OECD confidence interval should be interpreted cautiously given the small number of country clusters.

5.4 Model Diagnostics

Figure 5 presents residual diagnostics for the main FE specification. The residuals are approximately symmetric (skewness near zero) with modest excess kurtosis, indicating slightly heavy tails relative to a normal distribution. The Q-Q plot shows reasonable alignment with normality, with minor deviations in the tails. These diagnostics support the adequacy of the OLS-based FE specification for inference, though the normality test p -value should be interpreted cautiously given the large sample size ($N = 1,517$).

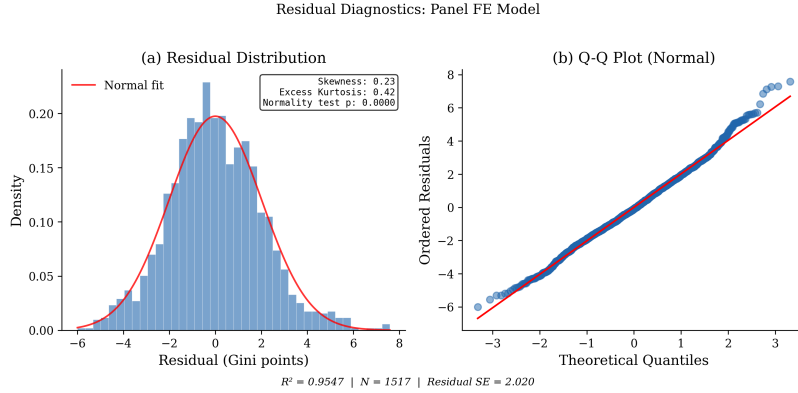


Figure 5: Residual diagnostics for the main panel FE specification. Left panel: residual distribution with normal density overlay. Right panel: Q-Q plot against normal quantiles. Summary statistics include skewness, excess kurtosis, and normality test p -value.

Notes: Residuals from the main specification (Column 2 of Table 1). The normality test p -value should be interpreted cautiously due to the large sample size.

6 Conclusion

This paper asks whether labor market institutions and welfare state generosity moderate the within-country relationship between trade openness and income inequality. Our answer is a qualified no. In a panel of 37 countries observed over 41 years, we find no statistically significant evidence that union density or social spending alters the trade-inequality gradient. The estimated marginal effects are modest in magnitude — shifting from weak to strong institutions changes the predicted effect of trade on the Gini by only 0.04 points — and the joint test of institutional moderation fails to reject the null. The unconditional trade-inequality relationship is itself small and imprecise, consistent with the existing consensus that no stable average effect exists (Lundberg and Squire, 2003; Milanovic, 2005; Jaumotte, Lall and Papageorgiou, 2013).

The Stolper-Samuelson pattern does, however, appear in the development-level heterogeneity: trade is associated with lower inequality in non-OECD countries and a near-zero effect in OECD countries, and this contrast is statistically significant. This finding suggests that the factor-abundance channel operates in the data, but it operates independently of the institutional channels we test. Institutions do not appear to be the mechanism driving the heterogeneous effects.

These results are bounded by three important limitations. First, the institutional variables available in cross-country panel data — union density and aggregate social spending — are coarse proxies for the complex institutional arrangements that shape labor markets and redistribution (Nickell and Layard, 1999). Bargaining centralization, coverage extension mechanisms, active labor market policies, and the composition of social spending may matter more than the aggregate measures used here. Second, slow institutional change over time limits the identifying variation available for interaction effects in a fixed effects framework; the within-country standard deviations of union density and social spending are small relative to their between-country variation. Third, the endogeneity of both trade and institutions remains a first-order concern that panel FE alone cannot fully address, and our robustness exercises do not include instrumental variables estimators that could strengthen causal interpretation (Frankel and Romer, 1999; Feyrer, 2009).

Future work should pursue three directions. First, richer institutional data — particularly measures of bargaining structure and welfare state composition that vary more within countries over time — could reveal moderation effects that aggregate proxies miss. Second, exploiting quasi-experimental variation in trade shocks (tariff reforms, trade agreement episodes) and institutional changes (labor reforms, welfare state retrenchment) would strengthen identification of the interaction effects. Third, mediation analyses that decompose the channels through which trade affects inequality — wage bargaining, social transfers, public goods provision — could clarify whether institutions operate on the distribution of market income, the redistributive system, or both.

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A Appendix: Robustness Checks

Baseline Specification without Interactions

The baseline panel FE specification without interaction terms (Column 1 of Table 1) yields a trade coefficient of -0.044 ($SE = 0.036$, $p = 0.218$). This unconditional estimate confirms that the within-country association between trade openness and net income inequality is negative but not statistically distinguishable from zero, consistent with the inconclusive findings of earlier work (Lundberg and Squire, 2003; Milanovic, 2005).

Bargaining Coverage as Alternative Labor Institution

When union density is replaced by collective bargaining coverage as the labor market institution measure, the results remain substantively unchanged. The trade coefficient is 0.022 ($SE = 0.051$, $p = 0.668$), the Trade $\times$ Bargaining interaction term is -0.007 ($SE = 0.071$, $p = 0.918$), and the Trade $\times$ Social interaction is -0.341 ($SE = 0.310$, $p = 0.271$). The model R^2 is 0.955 , comparable to the main specification. These results suggest that the null finding for institutional moderation is robust to the choice of labor institution proxy.

OECD and Non-OECD Subsample Estimates

The subsample analysis reveals meaningful heterogeneity by development level, as shown in Figure 4. The full set of subsample estimates is reported in Section 5. The contrast between OECD and non-OECD marginal effects is consistent with the Stolper-Samuelson prediction that trade reduces inequality in developing countries. The non-OECD effect is statistically significant at the 5% level despite the small number of country clusters (11 countries), providing suggestive evidence for the factor-abundance channel.